DevSecOps and Cloud Security

"Deliver resilient software capability at the speed of relevance"

Application Security Analyst and DevSecOps expert

- Web and mobile Developer
- +20 Application Security Evaluation
- SANS 540 DevSecOps and Cloud Security automation
- CTF Player/ Organizer
- I love microservices
- Cloud Native Application security advocate

What you will learn

All what you need to secure your CI/CD pipeline !

Agenda

Day 1 :

Attacking and Hardening the DevOps Toolchain

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Attacking and Hardening the DevOps Toolchain

Lab environment overview

Role	Systems(s)
Version Control	GitLab, CodeCommit
CI/CD	Jenkins, CodePipeline
Configuration manager	Ansible, CloudFormation
Container Execution	Docker, K8s
Secret Storage	Vault
Cloud Infrastructure	AWS

Lab setup summary

- Port conflict (GITLAB/JENKINS)
- HTTPS issue in GITLAB/JENKINS integration
- Authorizing local IP addresses in GitLab for WebHooks

Day 1 :Attacking and Hardening the DevOps Toolchain

- DevOps security challenges
- DevOps toolchain

DevOps success factors

CAMS (or CALMS) is a common lens for understanding DevOps and for driving DevOps change. Your organization succeeds when it reaches "CALMS":

- Culture: people come first
- Automation: rely on tools for efficiency and repeatability
- Lean: apply Lean engineering practices to continuously learn and improve
- Measurement: use data to drive decisions and improvements
- Sharing: share ideas, information, and goals across silos

DevOps Unicorns

The DNA of DevOps and DevSecOps comes from a few early leaders or"unicorns":

- Netflix: Undifferentiated heavy lifting in the cloud, Chaos Engineering
- Amazon: You Build It... You Run It, two-pizza teams dogfooding AWS
- Google: Site Reliability Engineering, fearless shared postmortems
- Twitter: Self-service security for developers, immediate feedback
- Etsy: Security in Continuous Deployment, "a Just Culture"

CALMS: Culture

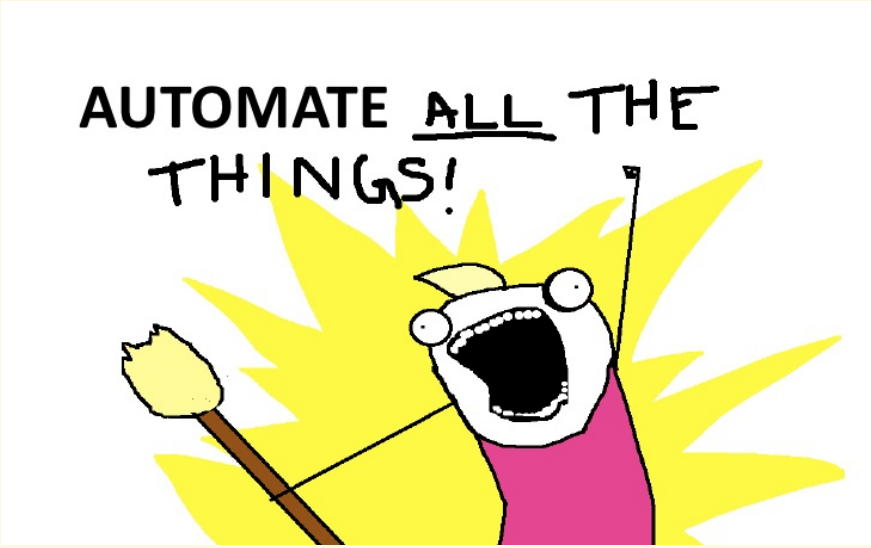
Signs of a culture that supports DevOps:

- Failures are accepted and used as learning opportunities.
  - Accept that failures can and will happen.
  - Utilize blameless postmortems to learn and improve from failures.
- Find ways to minimize impact or "blast radius."
- --Collaboration-- occurs widely across functional lines.
- Problem-solving: the problem is the enemy.
- Empower teams to take action on their own.
- Staff have a shared sense of ownership and accountability.
- --High levels of trust-- allow people to feel safe with each other.

CALMS: Automation

DevOps teams use programmable tools and cloud service APIs to:

- Take care of "undifferentiated heavy lifting" and repetitive work
- Continuously iterate and experiment in production
- Increase transparency
- Ensure consistency
- Prevent common mistakes
- Enable measurement and sharing
- Control operations at scale



CALMS: Automation "Everything as Code"

Get everything into code and check it into version control:

- Application source code/package dependencies
- Operations scripts
- Application configuration
- Tests and test scripts
- Build, deployment scripts, and runbooks
- Infrastructure provisioning and configuration
- Documentation
- Database definitions and change scripts
- Security/compliance policies
- Security tool configuration

CALMS: Lean engineering

Lean engineering improves efficiency, reduces friction, eliminates hand-offs and delays:

- Confront bureaucracy.
- Use Value Stream Mapping to identify waste, bottlenecks, and delays.
- Leverage Automation for routine work.
- Measurement: Use data to drive continuous improvements.
- Prioritize just in time-always work on what's important.

CALMS: Measurements

Measurements can help you to understand-and improve-how your teams work and where to automate:

1. Change frequency
2. Change failure/success rate
3. Correlation between change frequency and failure rate
4. MTTR recovery window instead of MTTF (time to failure)
5. Change lead time to deliver a change or fix, from check-in to deployment

CALMS: Measurements for DevSecOps

Collect security metrics along your delivery pipelines to assess how healthy your code is and how healthy your security program is.

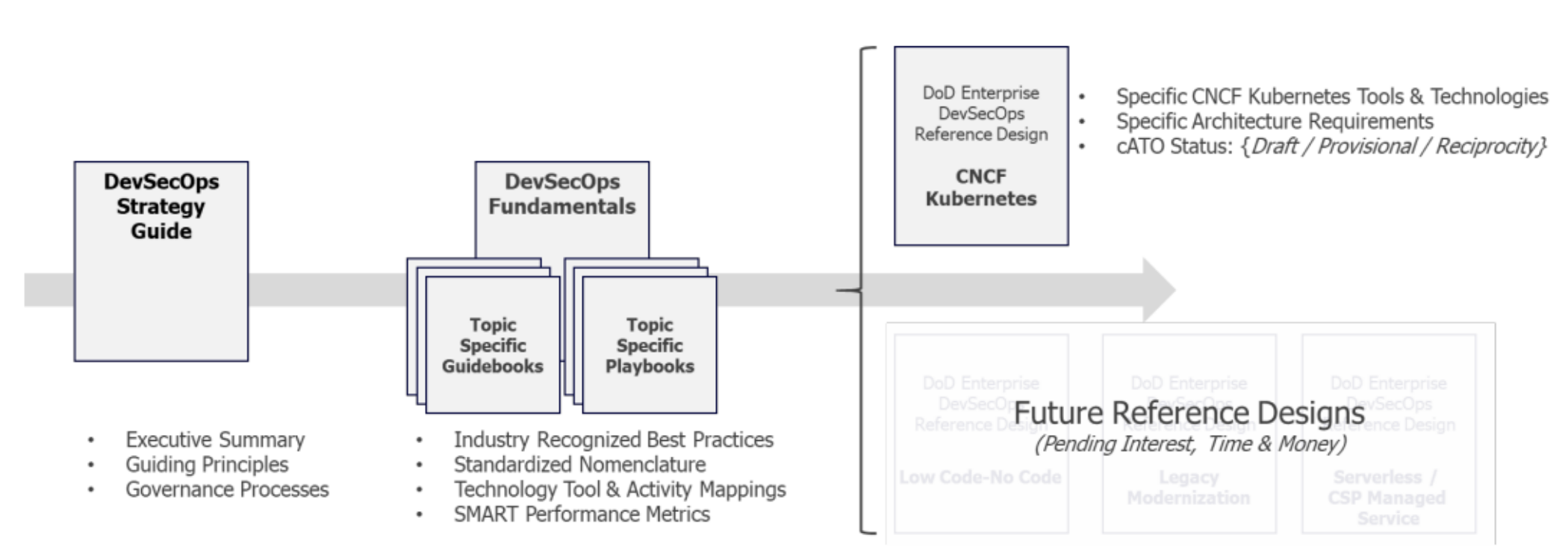
- Measure automated test coverage for high-risk code
- Track # of vulnerabilities found… and where they were found in the pipeline
- Track # of vulnerabilities fixed
- How long vulnerabilities remain open (window of exposure)
- Type of vulnerability (OWASP Top 10) for Root Cause Analysis
- Elapsed time for security testing-make feedback loops as short as possible
- False positives versus true positives-improve quality of feedback
- Vulnerability escape rate to production

CALMS: Sharing

Information should be shared freely across the organization.

- Everyone has access to all information that they need to do their job.
- Eliminate hand-offs where information/knowledge is often lost.
  Open up communications channels across teams/functional lines.
- Chat and ChatOps, shared code repos, backlog tools, dashboards, etc.
  Share goals and accountability across Dev, Ops, and Security.
- Everyone has a stake in success and preventing failures.
  Evangelism through conferences, blogs, articles, podcasts, etc.
- DevOps has an active, open community (DevOpsDays, open-source tools...).
- Vendors have latched on to this hyperactivity.

CALMS: US DOD case study



Security culture vs DevOps

DevOps culture conflicts with traditional security culture:

- Top-down risk management instead of team-based decision-making
- Need to know restrictions versus extended information sharing
- Zero failure versus fail fast and fail forward
- Limiting change: Security is always ready to say "No!"

Security Challenges in DevOps

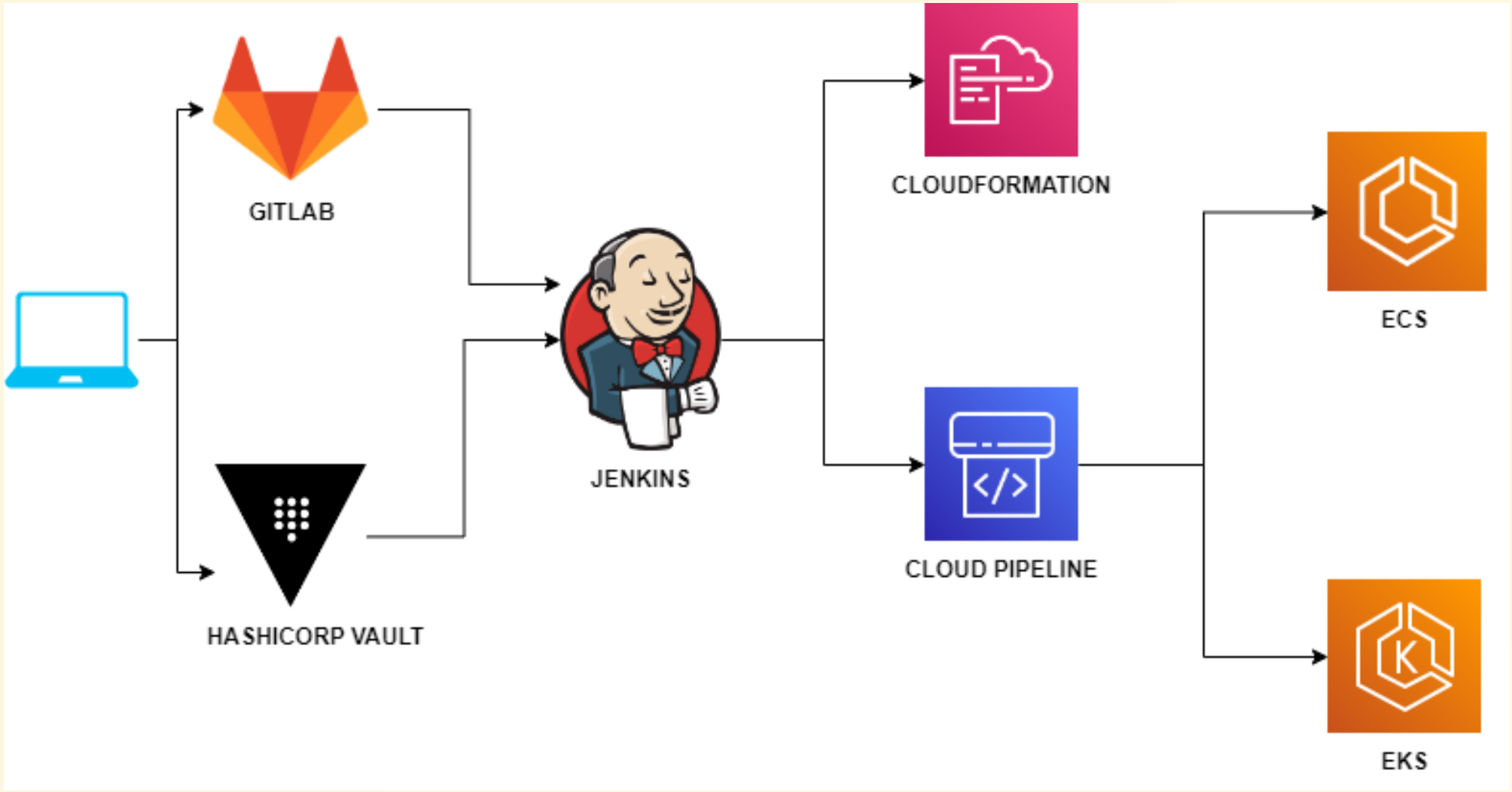
Powerful new technologies create opportunities for attackers-and new risks for organizations:

- Weaknesses in the DevOps toolchain can compromise the entire stack.
- Cloud platform misconfigurations can easily allow unauthorized access to data.
- Containers and orchestrators introduce a new attack surface, often not supported by traditional security scanners.
- Microservice-based architectures, new languages, and frameworks compound security guidelines.
- Delivery at the speed of DevOps requires enhanced detection and automated remediation.

Day 1 :Attacking and Hardening the DevOps Toolchain

DevOps toolchain

DevOps toolchain



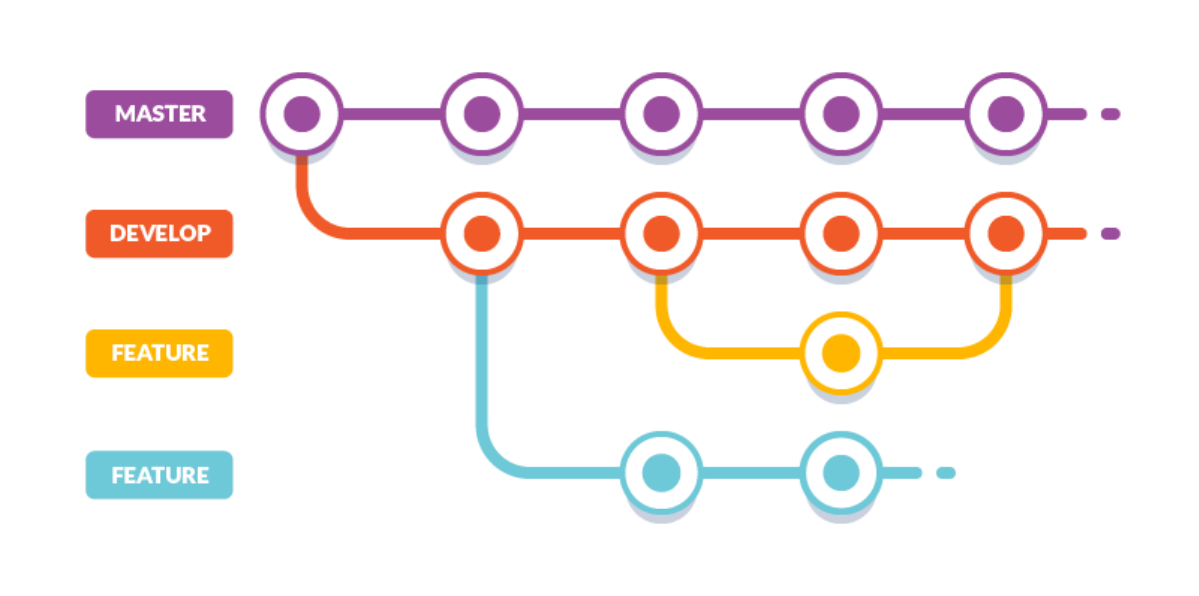
Version control

- The DevOps toolchain starts with pushing code to a version control system.
- The most common version control systems include:

GitFlow

Understanding GitFlow is important for security teams to:

- Contribute to development and operations workflows
- Prevent attackers from compromising the workflow



Continuous Integration

Approving and completing a merge request creates a commit on the "main" branch and automatically triggers a pipeline:

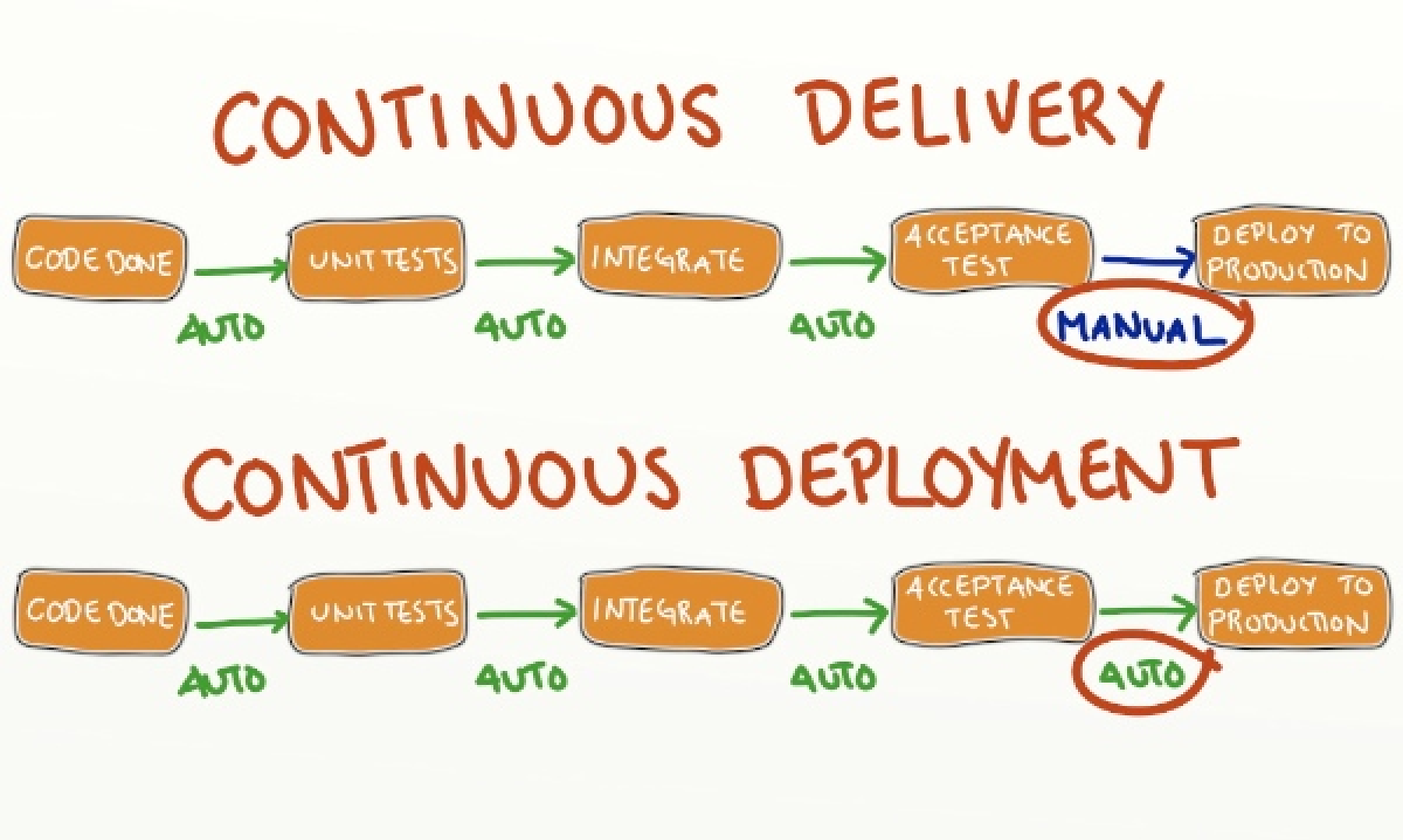
- Ensure that changes integrate successfully with the rest of the codebase
- Executes automated unit tests and other automated checks
  - Red: no other check-ins are allowed until the build is fixed
  - Green: creates build artifacts for next steps in CD pipeline
- Fast feedback - build and test steps need to run in a few minutes to encourage iterative development (small, frequent changes)

Continuous Delivery

Pipeline model and control framework extending Continuous Integration:

- It uses the latest good build from CI, packages for deployment, and release.
- Changes are automatically pushed to test/staging environments to conduct more realistic/comprehensive tests.
- It can insert manual reviews/testing/approvals between pipeline stages.
- Log steps and results to provide audit trail from check-in to deploy.
- Any failures will "stop the line."
- No additional changes can be accepted until the failure is corrected.
- This ensures that code is always ready to be deployed.
- Changes may be batched up before production release.

Continuous Delivery vs Continuous Deployement



Continuous Deployment

Continuous Deployment is how organizations like Netflix, Etsy, and Amazon push out changes n times per day/hour/minute:

- Changes are deployed directly and automatically to production using the CD pipeline once all tests/checks pass.
- Self-service - Changes are pushed to production by developers.
- Blue/Green Deployment - Deploy changes and switch between production environments using load balancing
- Canaries - Incremental deployment that stops and rolls back if errors occur
- A/B testing - Measuring the effect/acceptance of a change or new feature in production
- Dark Launching - Protect changes behind "feature switches"

CI/CD Systems

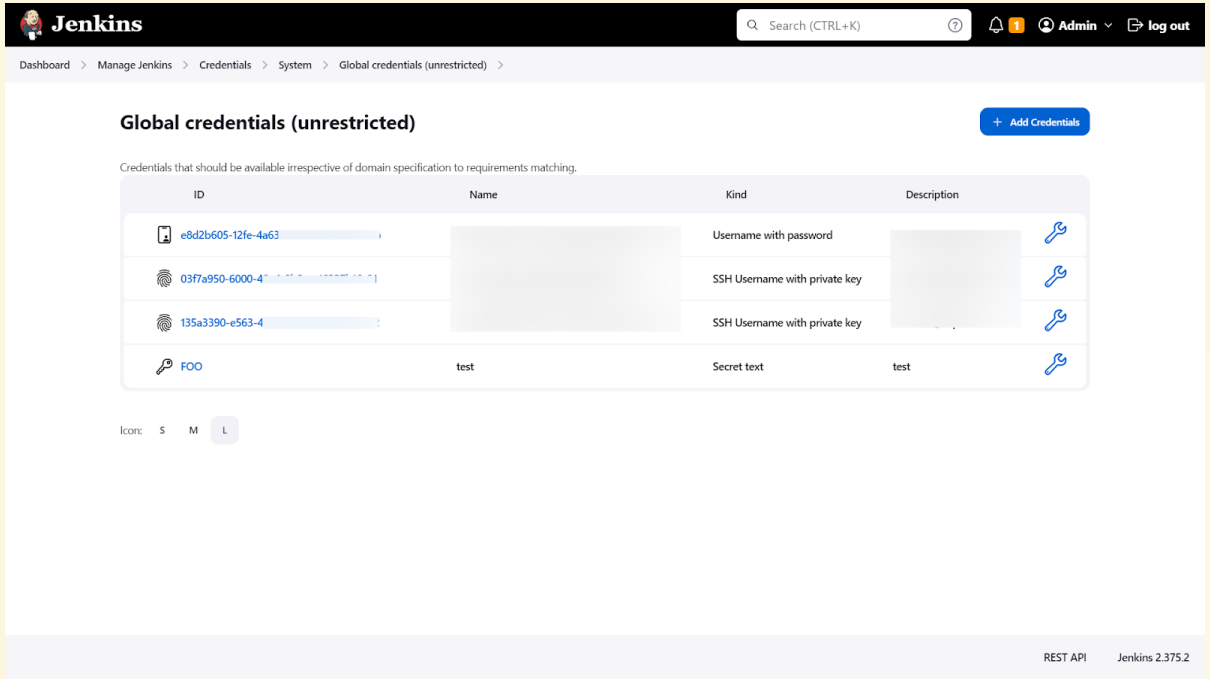
Version control push events on the develop/main branches trigger workflow pipelines for building, testing, and deploying the changes.

Jenkins pipeline as code declarative syntax

Jenkins pipeline building a docker image

```
pipeline {
    agent { dockerfile true }
    stages {
        stage('Test') {
            steps {
                sh 'node --version'
                sh 'svn --version'
            }
        }
    }
}
```

Jenkins credential manager




```
AWS_ACCESS_ID = ${env.AWS_ACCESS_ID}
AWS_SECRET = ${env.AWS_SECRET}
```

CI/CD Attacks

SOLARWINDS

CODECOV

CI/CD Security Risks

Top 10 **CI/CD Security** Risks

CICD-SEC-1	Insufficient Flow Control Mechanisms
CICD-SEC-2	Inadequate Identity and Access Management
CICD-SEC-3	Dependency Chain Abuse
CICD-SEC-4	Poisoned Pipeline Execution (PPE)
CICD-SEC-5	Insufficient PBAC (Pipeline-Based Access Controls)
CICD-SEC-6	Insufficient Credential Hygiene
CICD-SEC-7	Insecure System Configuration
CICD-SEC-8	Ungoverned Usage of 3rd Party Services
CICD-SEC-9	Improper Artifact Integrity Validation
CICD-SEC-10	Insufficient Logging and Visibility

OWASP

DevOps toolchain summary:

Security teams must understand how the CI/CD pipeline works, the tools involved, and how the teams use them before they start looking how and where to add security checks

Day 1 :Attacking and Hardening the DevOps Toolchain

Securing DevOps toolchain

DevOps / DevSecOps

Incremental, iterative development and rapid delivery in DevOps can significantly reduce this window of exposure:

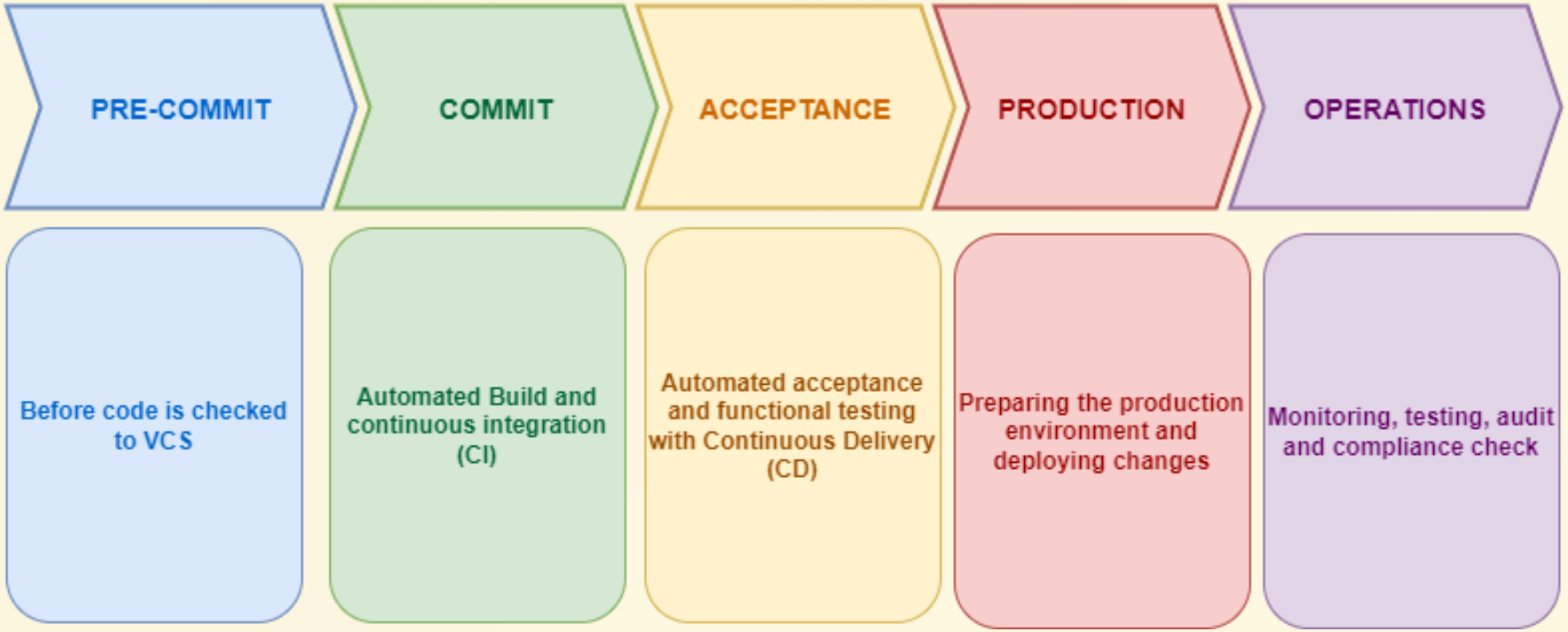
- DevOps teams follow "just-in-time prioritization"
- Automated testing cheap and safe to push out changes quickly.
- DevOps teams follow Lean techniques.
- Automated security checks in the CI/CD pipeline will catch many vulnerabilities.
- Security fixes are "standard changes".
- DevOps encourages developers and ops and security specialists to work together.

Security Champions: Breaking Down DevOps Silos

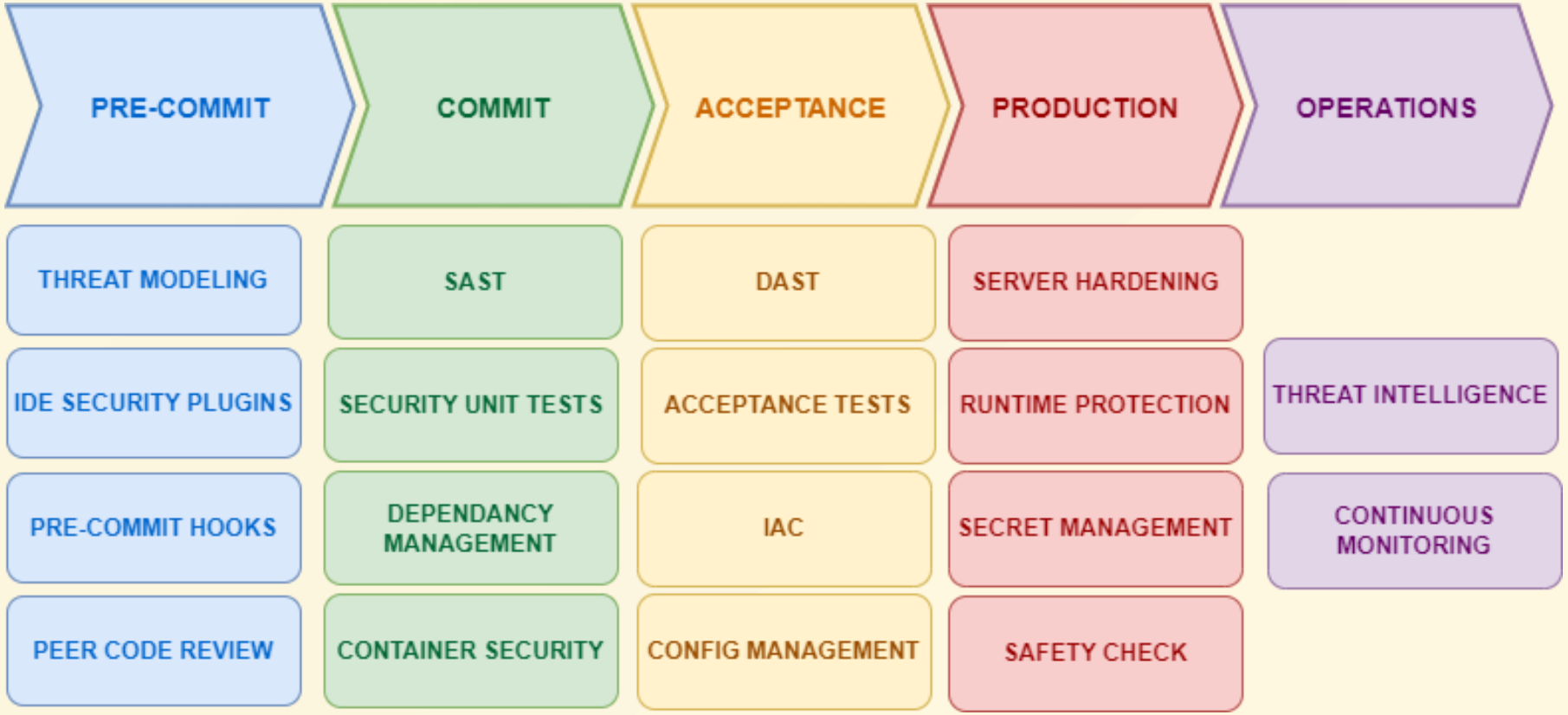
- Security experts embedded into the development organization
- Break down barriers between the central Security and DevOps teams
- Requires funding and support from upper management
- Provides several ancillary benefits:
  - The reputation of the company
  - Teaches valuable leadership skills
  - Increased employee retention

DevOps workflow

The DevOps workflow is based on five key phases:
-Manual work done before merging code into a delivery branch
-Automated Continuous Integration (CI) & Continuous Delivery (CD) process



DevOps workflow



DevOps workflow

RRA

- **Quick!** The RRA takes 30 to 60 minutes maximum.
- **Very high-level**. Details are for complete threat models. The RRA can become a complete threat model over time though!
- **Concise, readable**. Short and with clear risk levels.
- **Easy to update**. Can be run during any phase of the project development and continuously updated.
- **Informative**. Collects risk impact and a data dictionary. Also, collections information about how the service functions.
- **Let you know what to do**. The RRA includes the list of recommendations from the security team with a priority for each item.

Threat Modeling

Ask a few basic questions:

1. Are you changing the attack surface of the system in a material way? Are you adding a new API or opening a new port, adding a new user type/role, adding or changing a data store/sink, calling out to a new service or system?
2. Are you changing the technology stack? Are you adding or significantly upgrading a framework or important library or language or runtime component?
3. Are you changing security controls: authentication or session management code, cryptologic, access control code or rules, auditing logic...?
4. Are you adding confidential/sensitive data or changing data classifications?
5. Are you modifying high-risk code?

Store Markdown RRA and Threat Modeling in the repository /docs directory

Code Inventory

```
$ scc path/to/your/code
```

Language	Files	Lines	Code	Comments	Blanks
Go	112	50145	38876	5449	5838
Markdown	8	2228	1710	163	355
Shell	11	799	598	97	104
YAML	6	515	408	10	97
Total	137	53687	41670	5719	6394

Code Analysis: IDE Security Extensions

The VS Code Marketplace contains thousands of security-focused plugins:

- Semgrep
- Checkov/CFN Nag
- SonarLint/ ESLint
- Hadolint
- InSpec

Version Control Security

Version control systems and git provide workflow capabilities for locking down the remainder of the pre-commit security controls:

- Git Hooks
- Mandatory peer reviews
- Mandatory security reviews
- High risk code approvals

Git Commit Hooks

Run script automatically, checking for embedded secrets, code correctness at different point in the workflow :

- Local repository : pre-commit, commit, post-commit ...
- Remote repository: pre-receive, post-receive
- Implement team-wide workflow policies before CI

Manual Code Reviews

Take advantage of code review workflows for security reviews

- Security team sets policies and train developers on how to de security code reviews. Create a checklist for them.
- Look for problems that static analysis tools don't find.
- Tag High-risk code

Branch Protection

All version control providers provides a branch protection capability:

- Require merge requests to commit to the branch
- It defines the role that can complete a merge request.
- It prevents users to directly push to the main branch
- it enables the CodeOwners approval workflow

Code Owners

Inventory high risk code and identify groups and individuals responsible for approvals :

- Create CODEOWNERS file in the repository root directory
- Define directories or individual files in the CODEOWNERS file
- Require one or many approvers during the pull request to review changes
- The security team can participate in the GitFlow workflow

Code Owners

```
# This is a comment.
# Each line is a file pattern followed by one or more owners.
# Specify the owners for the entire repository
*       @all-developers
# Specify the owners for security-related files
/security/ @security
# Specify the owners for user experience-related files
/frontend/ @user-experience
# Example for specific files
README.md @all-developers
# Example for a specific directory
docs/ @all-developers
# Example for specific file types
*.js @all-developers
CODEOWNERS @security @admin
```

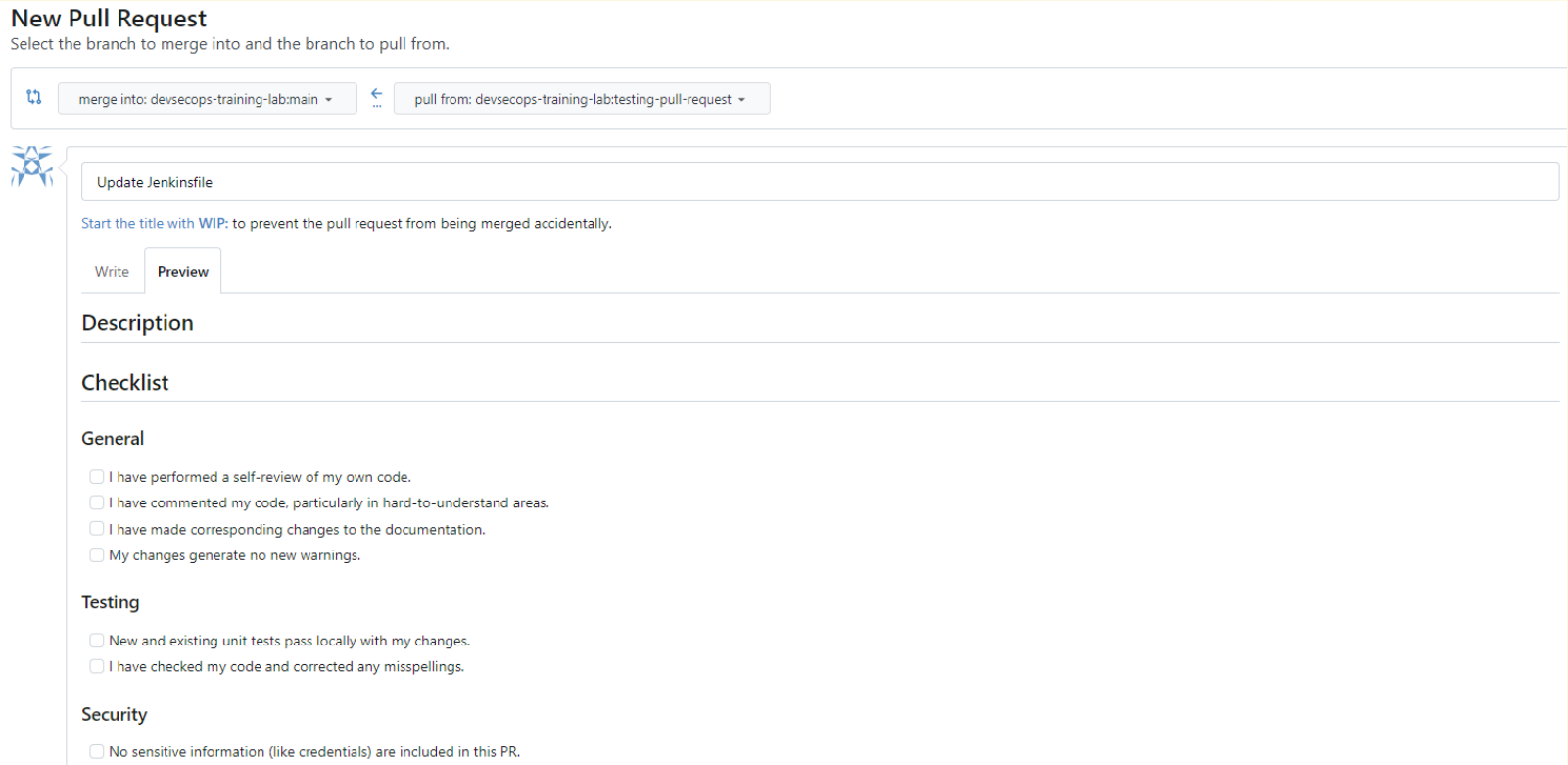
Detecting High Risk Code

Using unit tests to check if a high-risk code file changed.

```
describe('HighRiskCode Checksum Test', () => {
    const testCases = [
        { file: 'path/to/Jenkinsfile', checksum: '2bf33b66ddb07616f882ceed0718826af298a7' },
        { file: 'path/to/Dockerfile', checksum: 'fe83bf6f453698c5f78cab167bca14c72daf32c0' },
    ];
    testCases.forEach(({ file, checksum }) => {
        it(`should verify the checksum of ${file}`, () => {
            const alertSpy = sinon.spy(requestCodeReview);
            const actualChecksum = getChecksum(file);
            const match = checksum === actualChecksum;
            if (!match) {
                requestCodeReview(file);
            }
        });
    });
});
```

Security Checklist: Pull Request Templates

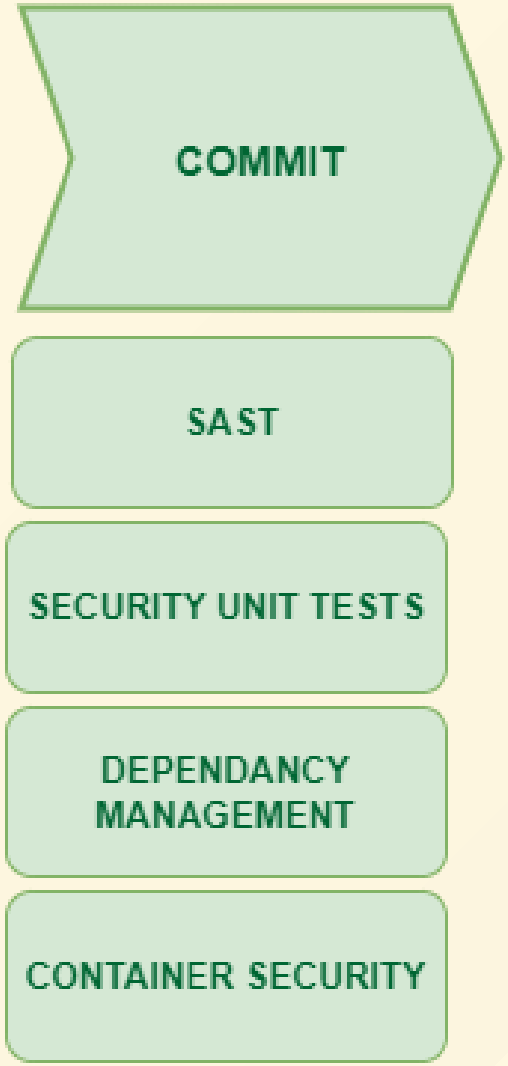
Markdown based templates for starting a pull request can include checklists, including security tasks



Summary

Secure your Version Control System

DevOps workflow - Security Controls



Static Analysis :

Provide fast and clear feedback on code commit:

- Run code scans in parallel with unit testing for speed.
- Run high-risk checks early in the pipeline to "fail fast."
- Do incremental scanning, if possible, as deep scanning takes too long for CI/CD, especially on large codebases,
- Set a max time limit with the dev team, alert them if the limit is breached and fix it.
- Return results directly to developers using their IDE or backlog list/tool.
- Run full/long-running scans out-of-band (nightly) or in the scanning

Minimize False Positives

Developers will tune out results if you don't tune out false positives.

- Carefully review rules to identify which checkers provide high-confidence results.
- Fail the pipeline if these checks fail.
- Turn off noisy, low-confidence checks in the pipeline-run them separately and review/qualify them manually. Feed real positive findings back to the development team's backlog.
- Periodically review and re-tune rules and configurations.
  Track and document tool configurations and decisions on which checkers or rules you disable (for compliance/governance).

SAST : Configuration management

IaC Technology	Static Code Analysis Tools
Ansible	KICS
CloudFormation	cfn_nag, Cfripper, Checkov, KICS
Terraform	Terrascan, Checkov, TFSec, KICS, ShiftLeft

SAST : Language Support

Type	Java	Node.js	Python	PHP
Code Quality	SonarQube, PMD	ESLint	Pylint, SonarQube	SonarQube, PHPStan, Psalm, Phan
Security Checker	Fortify, Checkmarx,	npm-audit, NSP	Bandit	RIPS
Bug Corrector	FindBugs	Flow	Pyflakes	Phan

SAST : Semgrep

Semgrep provides a lightweight, multi-language, extensible static analysis solution.

- Open-source
- Community driven rules
- Language support : Go, Java, JavaScript, Python ...
- Support automation !!!!!!!

SAST : Semgrep

```
semgrep --config "p/expressjs" --config custom_rules.yml /path/to/your/express/app

/path/to/your/express/app/index.js
  5:14  error  jwt-hardcoded-secret  Avoid hardcoding JWT secrets
  10:5  error  security-audit-rule   Some other security issue

2 issues
```

Semgrep : Custom rules

Custom rules can be added using the Semgrep playground to test the rule syntax :

```
rules:
- id: public-s3-policy-statement
  pattern: |
    {
      "Effect": "Allow",
      "Principal": "*",
      "Resource": [
        ..., "=~/arn:aws:s3.*/
        ", ...
      ],
      ...
    }
  message: Detected public S3
  bucket policy. This policy
  allows anyone to access
  severity: WARNING
  languages:
  - json
```

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "PublicReadGetObject",
            "Effect": "Allow",
            "Principal": "*",
            "Action": [
                "s3:GetObject"
            ],
            "Resource": [
                "arn:aws:s3:::example.com/*"
            ]
        },
        {
            "Sid": "CloudFront Origin Access Identity",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::cloudfront:user/CloudF
                ECLOUDFRONTOAI"
```

Dependencies : Component Analysis

Dependency Checker	Supported Languages
OWASP Dependency-Check	Java, JavaScript, .NET, Python, Ruby, PHP, Node.js
retire.js	JavaScript, Node.js
safety	Python
PHP Dependency Checker	PHP

Machine Readable report

Running security tools in CI/CD requires a supported machine-readable output format.

- xUnit/JUnit
  - Standard XML schema for reporting pass/fail unit test results
- Checkstyle
  - Standard XML schema for reporting static analysis results
- SARIF (Static Analysis Results Interchange Format)
  - JSON based schema primarily used for displaying results in GitHub
- JSON
  - Custom schemas are machine-readable, but you have work to do!

Containers Security

NIST SP 800-190

Containers will force you to use different security tools and practices:

- Adopt new technologies for writing secure Dockerfiles and scanning images.
- Understand how to sign and store images securely in a registry.
- Leverage container-optimized host operating systems (e.g., Google COS) to reduce attack surface.
- Obtain container-specific runtime security appliances that manage dynamic and ephemeral containerized environments. Perimeter-based runtime defenses like IDS/IPS/WAF do not have visibility inside container networks.

Container Security life cycle

Security must shift left and consistent policies are implemented along each step of the container life cycle:

1- Pre-commit
2- Version Control
3- CI/CD
4- Container Registry
5- Container Orchestrator

Dockerfile Commands

Command	Description
`FROM`	Specifies the base image to use for the Docker image.
`LABEL`	Adds metadata to the image, such as a maintainer or version.
`RUN`	Executes a command in a new layer on top of the current image and commits the results.
`CMD`	Provides the default command to run when the container starts. Can be overridden with `docker run`.
`ENTRYPOINT`	Configures a container to run as an executable. Unlike `CMD`, it is not overridden by the `docker run` command.
`WORKDIR`	Sets the working directory for any `RUN`, `CMD`, `ENTRYPOINT`, `COPY`, and `ADD` instructions that follow it.
`COPY`	Copies new files or directories from `<src>` and adds them to the file system of the container at the path `<dest>`.
`ADD`	Similar to `COPY` but can also extract local tar archives and download remote files.
`ENV`	Sets an environment variable.
`EXPOSE`	Informs Docker that the container listens on the specified network ports at runtime.
`USER`	Sets the username or UID to use when running the image and for any `RUN`, `CMD`, and `ENTRYPOINT` instructions that follow it.
`VOLUME`	Creates a mount point with the specified path and marks it as holding externally mounted volumes from native host or other containers.
`ARG`	Defines a variable that users can pass at build-time to the builder with the `docker build` command.

Hunt The Bug

```
FROM mcr.rnicrosoft.com/dotnet/core/sdk:2.2
ARG ENVIRONMENT=prod
ENV ASPNETCORE ENVIRONMENT=$(ENVIRONMENT}
ENV CERTIFICATE_PASSPHRASE-mycertpassphrase
RUN apt-get update && apt-get install -y libss1-dev=1.1*
COPL cert.conf -/cert.conf
RUN openssl reg -x509 -nodes -days 3650 -newkey rsa:4096 -keyout app.key
-out app.ert -config cert.cont -passin pass:$(CERTIFICATE_PASSPHRASE)
RUN openssl pkcsl2 -export -out app.pfx -inkey app.key -in app.crt
-passout pass: $(CERTIFICATE_PASSPHRASE]
USER root
WORKDIR /WWW
ENTRYPOINT ["dotnet", "Sans. CreditUnion API.dil"1]
```

Container Security : Pre-commit

Container security starts upfront the pre-commit phase with policies:

- SAST for Dockerfile misconfiguration
- Locking down the base image supply chain
- Installing approved binaries inside a base image
- Using multi-stage builds to create minimalistic images
- Passing build time secrets to image build commands

SAST for Dockerfiles

Tool	Description
Hadolint	A linter for Dockerfiles that helps you build best practice Docker images by providing tips.
KICS	An open-source tool that finds security vulnerabilities, compliance issues, and misconfigurations in IaC.
Dockerfilelint	A linter for Dockerfiles to ensure adherence to best practices.
dockerfile-lint	Another linter for Dockerfiles focusing on best practices and potential issues.

Hadolint

Hadolint parses Dockerfiles into Abstract Syntax Trees (AST) and runs rules:

- Supports an approved list of trusted registries
- Suppress false positives via inline comments or a configuration file
- Export formats include checkstyle, sarif, and json

```
$ hadolint --trusted-regestry gcr.io ./Dockerfile
Dockerfile:4 DL3008 warning: Pin versions in apt-get install. Instead of `apt-get install -y python3` use `apt-get install -y python3=<version>`
Dockerfile:7 DL3018 info: Pin versions in pip. Instead of `pip install -r requirements.txt` use `pip install -r requirements.txt && pip freeze > requirements.txt`
Dockerfile:9 DL4000 warning: MAINTAINER is deprecated
Dockerfile:10 DL3020 error: Use COPY instead of ADD for files and folders
Dockerfile:14 DL3025 info: Use arguments JSON notation for CMD and ENTRYPOINT commands
```

Supply chain hardening

MITRE ATT&CK Containers T1525: Implant Internal Image

- Images from public registries may contain vulnerabilities or malware, easy and common attack vector
- Mitigations:
  - Building inventory of approved base images
  - Downloading base images from trusted suppliers
  - Scanning base images for vulnerabilities
  - Creating a private container registry
  - Signing custom images and storing in a private registry

Container image trusted suppliers

- Docker hub (Verified Publisher)
- Platform One Iron Bank
- MCR
- ECR Gallery

Conftest - Trusted supplier policy

The big difference between Contest and a static analysis engine like hadolunt is the ability to wnte custom tests using the Open Policy Agent (OPA - pronounced oh-pa) Rego query language. The example above shows how to create a trusted images Rego policy.

```
package main
trusted_suppliers = ["ubuntu:20.04", "alpine:3.13", "node:14-alpine"]
deny[msg] {
    input[i].Cmd == "from"
    image := input[i].Value
    not image in trusted_suppliers
    msg = sprintf("The image '%s' is not from a trusted supplier.", [image])
}
```

Conftest - Trusted supplier policy

Pre-commit hooks can validate conftest policies for trusted base images and package install commands:

```
$ conftest test --namespace image, install Dockerfile
```

- Failing policies are shown in the console output
- Optional output formats for table, junit and json

```
FAIL - Dockerfile - image - [untrusted_supplier] Untrusted supplier identified: mer.rnicrosoft.com/dotnet/core/sdk:2.2
FAIL - Dockerfile - install - [install_command] Disallowed command found: ["apt-get update &s apt-get install -y libss1-dev=1.1**]
2 tests, 0 passed, 0 warnings, 2 failures, 0 exceptions
```

Minimal image with multi-stage Builds

Multi-stage builds use multiple FROM statements to differentiate "development" from "runtime" base images.

- Stage 1 uses a base image with an SDK to build the application (1.5GB)
- Stage 2 copies the binaries into an Alpine runtime image (120MB)

```
FROM mcr.microsoft.com/dotnet/nightly/sdk:6.0 AS build
COPY ./Web/  ./Web/
RUN dotnet publish "./Web/Website.csproj" -c Release -o /app/bin
FROM mcr.microsoft.com/dotnet/nightly/runtine:6.0.2-alpine3.15
WORKDIR /www
COPY --fron-build /app/bin ./ ENTRYPOINI ["dotnet", "Website.dll"]
```

Configure Build Time Secrets

Docker BuildKit supports build secrets (--secrets) to help developers avoid hard-coding secrets in Dockerfiles and environment variables:

```
COPY cert.conf /cert.conf
RUN --mount-type-secret, id-passphrase openssl req -x509 -nodes -days 3650 - newkey rsa: 4096 -keyout app.key -out app.ert -config cert.conf
-passin file: ///run/secrets/passphrase
RUN --mount-type-secret, id-passphrase openssl pkes12 -export -out app.pfx - inkey app.key -in app.ert -passout file: ///run/secrets/passphrase
```

Running the docker build command, setting the tag, and configuring a build time secret:

```
echo $ (CERTIFICATE_PASSPHRASE) > passphrase
docker build --tag dntools/base: insecure --secret id-passphrase, sro-passphrase --file Dockertile
```

Docker scan

- Scan images and Dockerfiles with docker scan CLI (Snyk) before committing Dockerfile
- Requires Docker Hub account to run scan locally

```
$ docker scan --file./Dockerfile
X Critical severity vuinerability found in Curl/1ibcur13
Description: Double Free
Into: https: //snyk.10/vuln/SNYK-DEBIAN9-CURI-466508
```

Container Security - CI/CD

Merge requests triggering CI/CD workflows responsible for building and releasing container images must certify the image:

- Enforce pre-commit controls and validate results.
- Build a release candidate image and perform in-depth scanning for vulnerabilities.
- Sign the release candidate image, tagging for production, and pushing to the container registry.

Open Source Container Scanning Tools

- Anchore
- Docker Bench
- OpenSCAP plugin
- Trivy

Trivy Image Scanning

- Scannning the container image in CI/CD
- Output format : table, json, sarif ...

Cosign Container Image Scanning

- Cosign support artifact signings
- Signing key storage in AWS, Azure, GCP, VAULT, and on premise

```
$ docker push 123456789012.dkr.ecr.us-west-2.amazonaws.com/api-image:1.4
$ cosign sign --key awskms:///alias/dm/cosign 123456789012.dr.ecr.us-west-2.amazonaws.com/api-image:1.4

1.4: digest: sha256:b30e9b9aße0e29d1852c07825dcca52d51ad58920849851747c64a5379bb1a95 size: 1793
Pushing signature to: 123456789012.dkr.ecr.us-west-2.amazonaws.com/api-image
```

Container Security - Registryg

Private container registries storing production images are the bridge between development and operations:

- They store image signature metadata for validation.
- They enforce strict access control on registry read and write operations.
- They are continuously scanning registries for new vulnerabilities and sending notifications.

AWS ECR - Best practices

AWS Elastic Container Registry (ECR) security best practices:

- Isolate container registries in a dedicated DevOps account
- Restrict access using resource-based and identity-based IAM policies for cross-account access
- Protect production images from corruption or deletion with immutable tagging
- Enable basic container image scanning

Container Security - Runtime

Container runtimes and orchestrators are responsible for pulling registry images and managing the workloads:

- Review security hardening guides for the selected runtime
- Admission controllers validate image signatures and security
- requirements
- Runtime security appliances inspect network traffic, monitor lateral communications, quarantine containers, and run vulnerability scans

kubernetes Security/Insecurity

Insecure Workload Configurations
Supply Chain Vulnerabilities
Overly Permissive RBAC
Policy Enforcement
Inadequate Logging
Broken Authentication
Network Segmentation
Vulnerable Components

Istio

Istio is an open source service mesh that layers transparently onto existing distributed applications. Istio's powerful features provide a uniform and more efficient way to secure, connect, and monitor services. Istio is the path to load balancing, service-to-service authentication, and monitoring – with few or no service code changes

Istio gives you

Secure service-to-service communication in a cluster with mutual TLS encryption, strong identity-based authentication and authorization
Automatic load balancing for HTTP, gRPC, WebSocket, and TCP traffic
Fine-grained control of traffic behavior with rich routing rules, retries, failovers, and fault injection
A pluggable policy layer and configuration API supporting access controls, rate limits and quotas
Automatic metrics, logs, and traces for all traffic within a cluster, including cluster ingress and egress

Effects of crime

1- Workload disruptions
2- Resource Consumption issues
3- Traffic issues : latency
4- Security issues

Ambiant Mesh : 2022

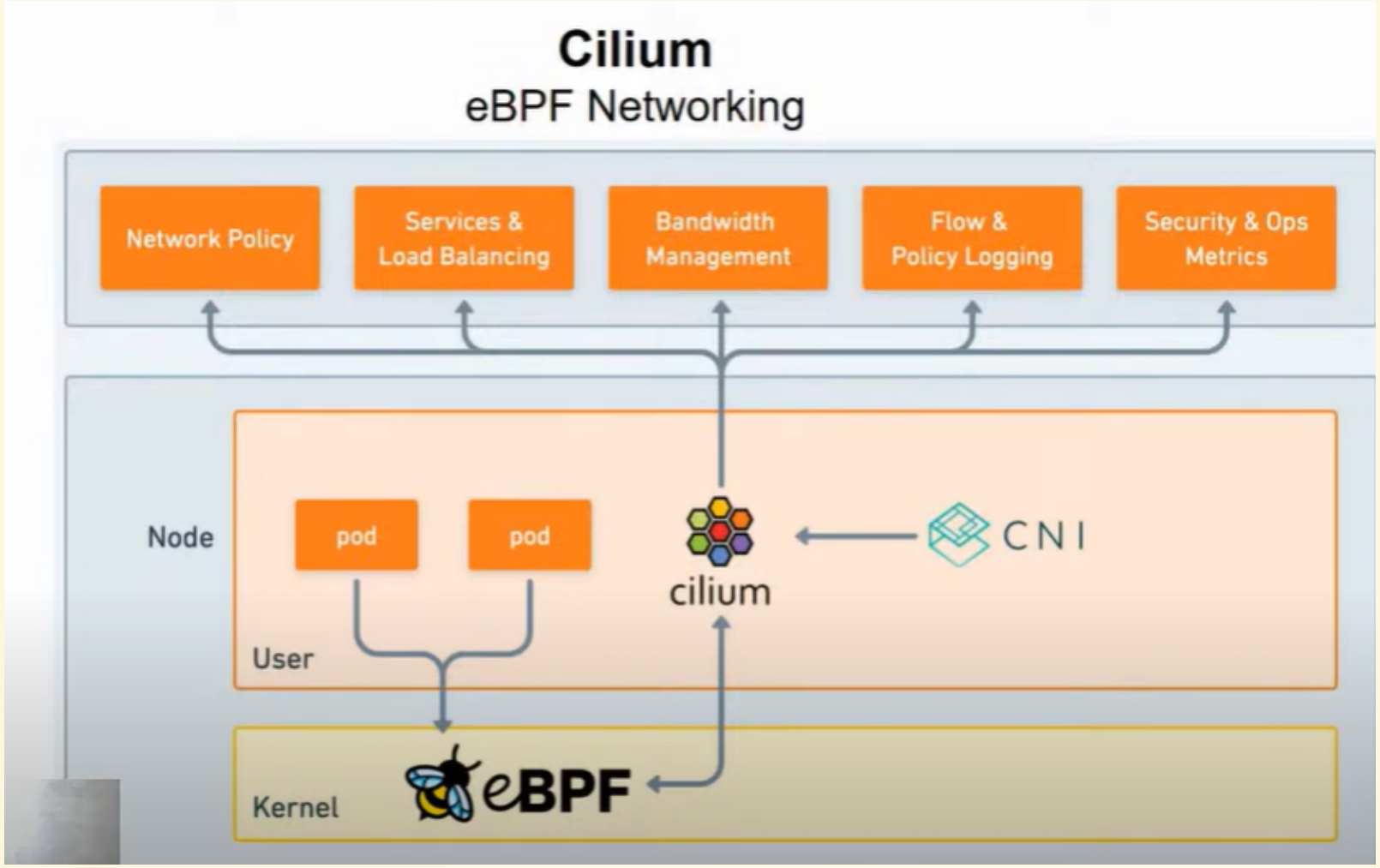
level 4 communication : TCP routing
Security : mTLS tunneling, ZTunnel
Observability : TCP metrics and logging

Resource efficiency
Optimal scaling of L7 proxies
Improved performance
Increased Security

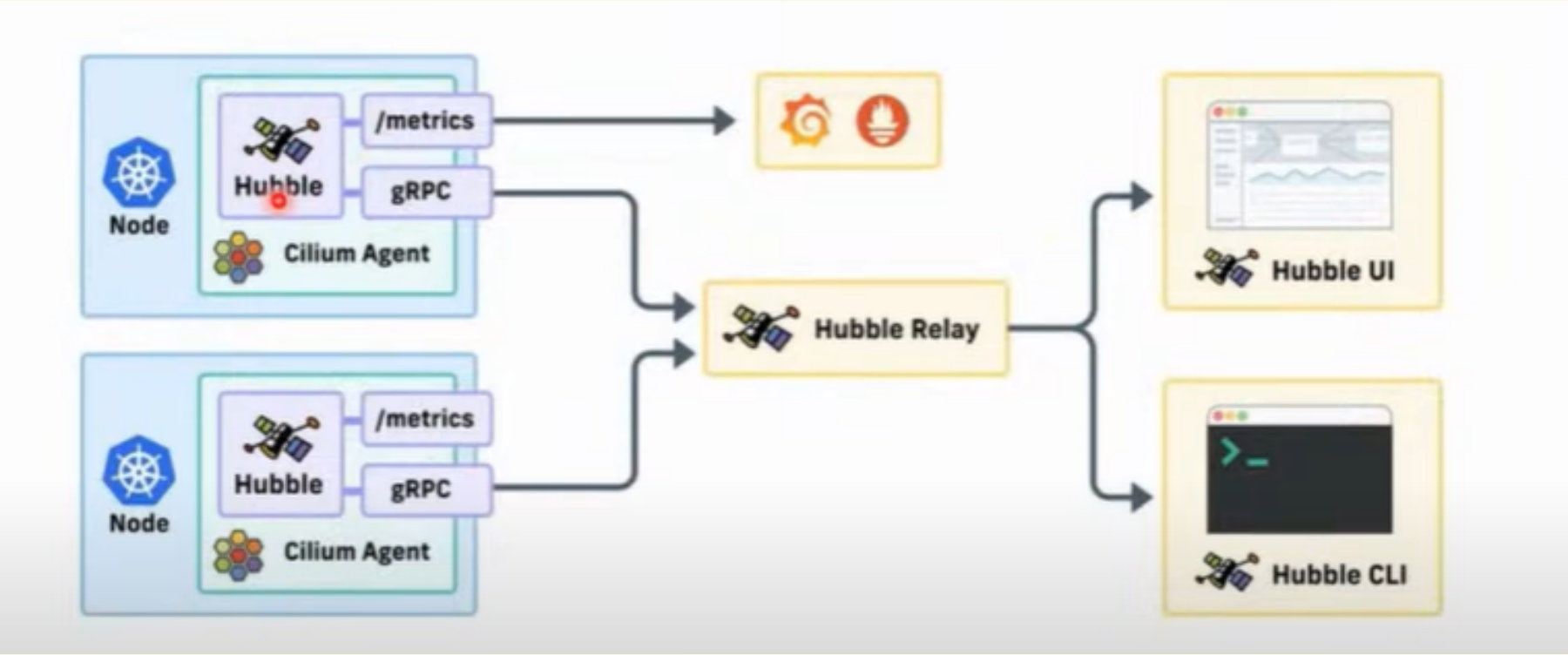
eBPF

Cilium eBPF based service Mesh

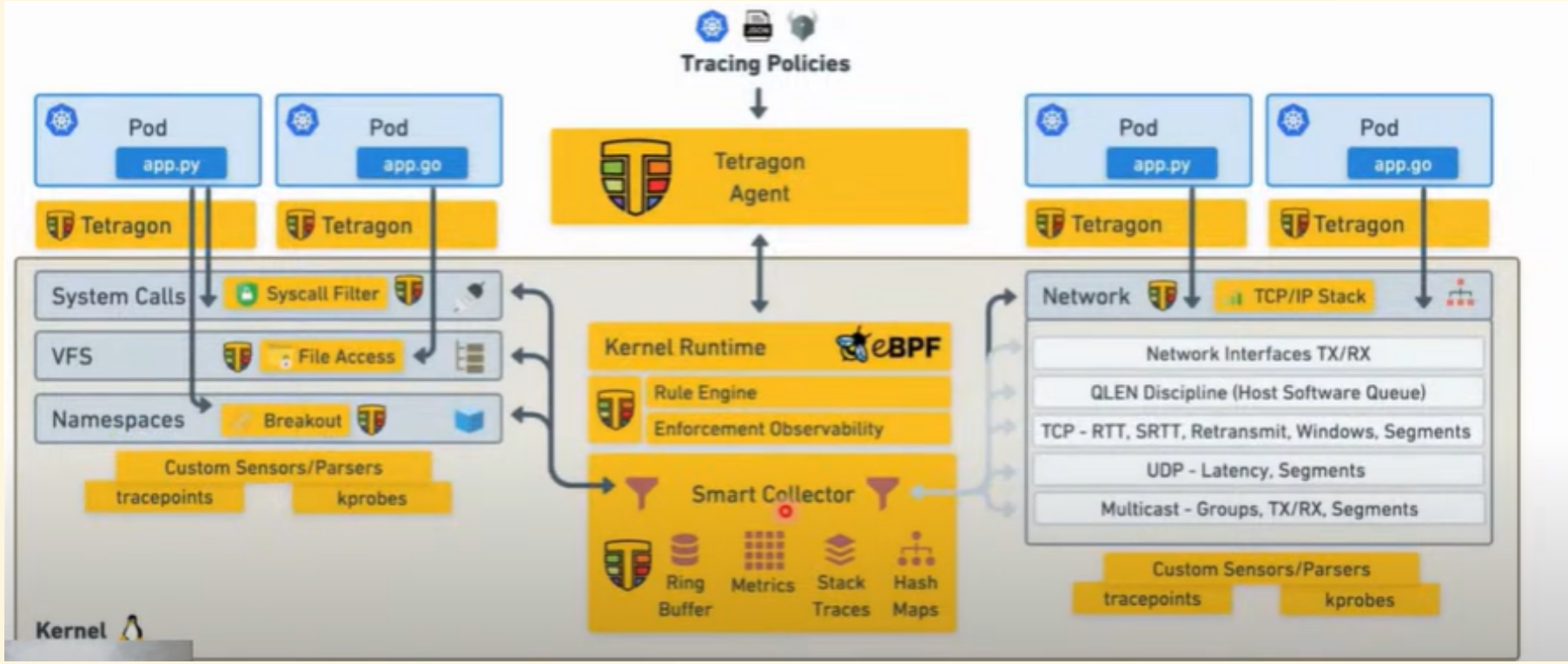
Cilium Service Mesh



Cilium Service Mesh



Cilium Service Mesh



Infra Structure As Code (IaC)

Security and best practices

IaC Pipeline

With the Infrastructure as Code (IaC) templates written, all configuration changes should go through the IaC CI/CD pipeline. This requires the engineer to check code into version control using a feature branch to create a pull request that will then be merged back into the main branch. Once this occurs your Infrastructure as Code pipeline is kicked off.

- Develop Environment
- Stage Environment
- Production Environment

IaC Security

Infrastructure as Code (IaC) deployments introduce new security concerns:

- Templates provide a blueprint of the entire cloud environment.
- Secrets are everywhere.
  - Service account keys, SSH keys, API keys, Terraform state file
- The Least privilege is hard.
  - Service accounts deploying templates require elevated permissions
- There are more supply chain attacks.
  - Terraform modules and nested CloudFormation templates
- Code analysis (SAST) requires a different toolchain.

IaC Security

- Understand The Configuration (RRA)
- Participate in peer reviews
- Integrate code analysis tools
- Set up security rules

AWS EC2

```
Resources:
  MyEC2Instance:
    Type: 'AWS::EC2::Instance'
    Properties:
      InstanceType: t2.micro
      SecurityGroups:
        - !Ref MySecurityGroup
      ImageId: ami-0c55b159cbfafe1f0
```

AWS EC2

```
MySecurityGroup:
  Type: 'AWS::EC2::SecurityGroup'
  Properties:
    GroupDescription: 'Allow HTTP and HTTPS traffic'
    SecurityGroupIngress:
      - IpProtocol: tcp
        FromPort: 80
        ToPort: 80
        CidrIp: 0.0.0.0/0
      - IpProtocol: tcp
        FromPort: 443
        ToPort: 443
        CidrIp: 0.0.0.0/0
```

IaC Security Scanning Tools

- Terrascan
- Semgrep
- Kicks
- Easy_infra
- checkov
- cfn_nag

checkov

Checkov is a Python command-line package:

- Support Terraform, CloudFormation,K8s, Dockerfile.
- Thousands of security policies across AWS, Azure and GCP.

Option	Description	Usage Example
`-d`	Directory to scan for infrastructure as code files	`checkov -d /path/to/iac`
`-f`	Specific file to scan	`checkov -f /path/to/iac/file.yaml`
`-o`	Output format (can be `cli`, `json`, `junitxml`, etc.)	`checkov -o json -d /path/to/iac`
`-c`	Path to Checkov configuration file	`checkov -c /path/to/checkov/config.yaml`
`--framework`	Specify the framework to scan (e.g., `terraform`, `cloudformation`, `kubernetes`, etc.)	`checkov --framework terraform -d /path/to/iac`

Checkov running the scan

Running Checkov on the templates' directory

```
checkov -directory ./templates --framewok terraform --output json > checkov.json
```


```
{
  "summary": {
    "passed": 2,
    "failed": 2,
  },
  "results": [
    {
      "check_id": "BC_AZR_NETWORKING_3",
      "check_name": "Ensure that Network Security Group inbound rule does not allow ALL ports from ALL traffic",
      "check_result": {
        "result": "FAILED"
      },
      "resource": "azurerm_network_security_rule.example"
```

checkov False positives

Suppressing the CKV2-AZURE-10 false positive using the checkov:skip comment:

```
ressource "azureem_virtual_machine" "bastion_vm"{
  #checkov:skip=CKV2_AZURE_10: Antivirus and patching is used with an EDR
}
```

CloudFormation Nag

cfn_nag is a CloudFormation template checker:

- Rule support includes IAM, security groups, logging, encryption ...
- Accept -i for the input
- Accept -o for the output that could be text or json

cfn_nag , False Positives

Suppressing the W2 and W9 cfn_nag false positives using the metadata attribute:

```
SecurityGroupPublic:
  Type: AWS::EC2:: SecurityGroup
  Metadata:
    cfn_nag:
      rules_to_suppress:
        - id: W9 reason: "This policy has been approved by security. (REF - 06152018]
        - id: W2 reason: "This policy has been  approved by security. (REF - 06152018]'
Properties:
```

Configuration Management as Code

Infrastructure as Code is a new way of working for many System administrators. It is less about system administration and more about software engineering:

- Establishing and following coding standards and patterns and defining and enforcing security and compliance policies in code.
- Organizing code into modules and creating reusable functions and templates.
- Implementing static analysis/linting tools to catch common mistakes and enforce good coding practices/style.
- Checking all code/configuration into version control.
- Establishing team disciplines of code reviews and refactoring.
- Writing automated tests to prove that the changes are implemented correctly-including Test-Driven
  Development (TDD) where tests are written before the code is changed (red/green).

Configuration Management Tooling

Declarative

" What ?", Desired State, easy verification, Puppet, CFEngine

Procedural

"How ?", Detailed steps, Easy to use and understand, End state not captured, Ansible, Chef

Configuration Management as Code Hardening

Follow basic good practices in setting up configurations and take advantage of hardening templates/recipes and examples.

- dev-sec.io open-source hardening framework
- CIS benchmarks
- DoD STIG templates
- SIMP Project from NSA (https://www.simp-project.com/)

Always carefully review open-source templates/modules before use.

Configuration Management - Test System

- Phoenix test Servers
- Vagrant can be a rapid and simple tool to spin temporary disposable VM sandbox for testing config changes
- After testing, tear down and clean up the test runtime.
- Reduce the ATTACK surface

Building Gold Images with Packer

Packer is an open-source tool for creating machine images on different platforms with a common configuration:

- Source images from Amazon EC2, Azure Virtual Machines, Google Compute Engine, Docker, VMWare, Virtual Box
- Apply Ansible, Puppet, or Chef configurations
- Publish a custom virtual machine image to the target platform
- Command-line driven and built for CI/CD

Ansible security

Automated testing tools for Ansible:

1. Run ansible-lint before committing code to source repository.
2. Run ansible-test for unit and integration tests suites.
3. Run Checkmarx's KICS scanner to check for dangerous Ansible code.
4. Run InSpec stories against test VM to validate final configuration

InSpec : Automationg Compliance Checking

Open-source language to write compliance checks that auditors can understand.

- Works like Serverspec: test the actual configuration against expected results + report deviations
- Tests include severity/risk level and description metadata to match against compliance checklist items or regulatory policies
- Check resources on Linux, Unix,Windows, AWS, Azure, GCP, VMware InSpec tests become your compliance requirements.

Make your own harden images

Jenkins + Packer + Ansible + Inspec